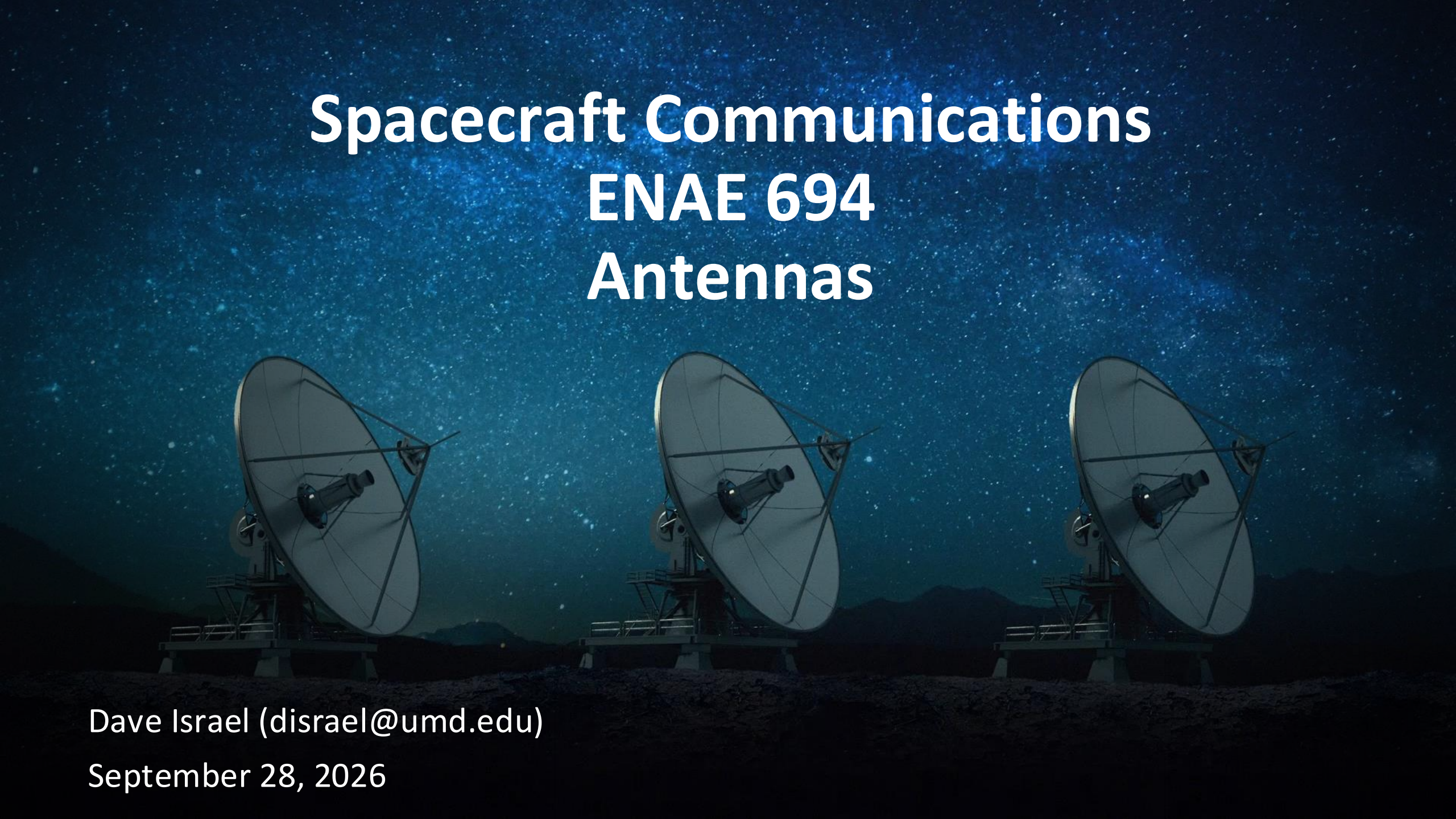
Spacecraft Communications
ENAE 694
Antennas



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September 28, 2026

Tonight's Agenda

1. Homework Review
2. Antenna Tour
3. Gain, Beamwidth, and Polarization
4. Coverage, Steering, and Mission Trades

Homework 2: Slant Range

A spacecraft is in a circular orbit 500 km above Earth. Calculate its slant range from a ground station when its elevation angle is 10°.

Earth Radius = 6,378 km Altitude = 500 km Elevation = 10°

$$R = \sqrt{((R_e + h)^2 - R_e^2 \cos^2 e)} - R_e \sin e$$

R ≈ 1,695 km

Why does a higher minimum elevation reduce maximum path loss?
What happens to contact time?

Homework 2: Voyager 2

2. Voyager 2 Received Power • 60 Points

Calculate total received signal power for Voyager 2's X-band link to DSS-43, the 70 m antenna in Canberra. Use the updated range below with radio parameters from JPL's January 1, 1996 budget [1].

Range: 21,476,670,660 km; frequency: 8.415 GHz; transmit power: 12.3 W; transmit antenna gain: 48.20 dBi; receive antenna gain: 74.01 dBi. Use 0 dB transmit feed loss and 0.42 dB other losses (pointing, atmosphere, and polarization combined).

a. Calculate [EIRP] in dBW. [20 points]

b. Calculate free-space path loss [L_p] in dB. [20 points]

c. Calculate received power [P_r] in dBW. Round your results to one decimal place. [20 points]

Homework 2: Voyager 2

Quantity	Calculation	Result
[EIRP]	$10 \log_{10}(12.3) + 48.20$	59.10 dBW
$[L_p]$	$32.45 + 20 \log_{10} R + 20 \log_{10} f$	317.59 dB
$[P_r]$	$59.10 - 317.59 + 74.01 - 0.42$	−184.90 dBW

R = 21,476,670,660,000 m f = 8.415 GHz

Is received power alone enough to predict data recovery?

Homework uses the supplied September distance with historical radio parameters [1].

Quantities and Their Units

Quantity	Linear Form	Logarithmic Form
Power or EIRP	W	dBW or dBm
Antenna Gain	Ratio to Isotropic	dBi
Loss	Power Ratio ≥ 1	dB, Subtracted
Power Flux Density, S	W/m^2	Linear Form Today
Effective Area, A_e	m^2	Linear Form Today

Course notation: brackets denote logarithmic quantities. 0 dBW = 1 W = 30 dBm.

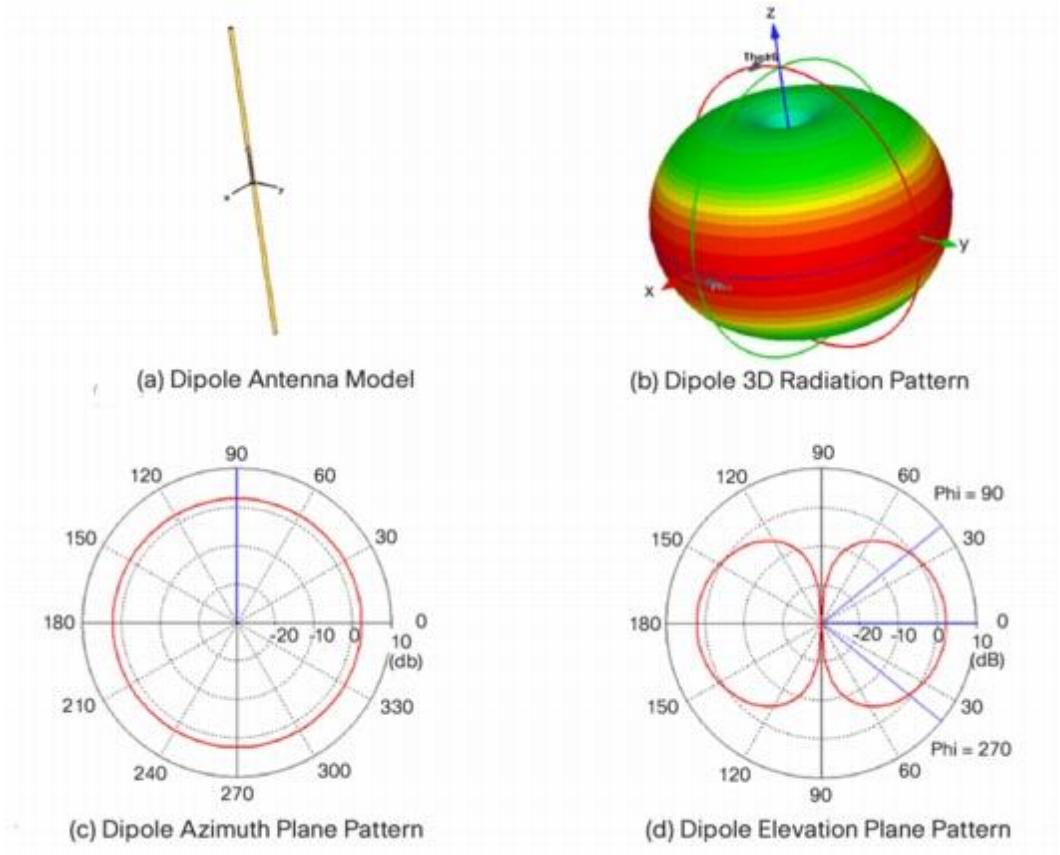
What Does Antenna Gain Mean?

How can an antenna increase the signal without increasing transmitter power?

Gain describes radiation in a particular direction,
relative to a lossless isotropic antenna with the same input power.

The engineering question: where does that stronger signal need to go?

Isotropic and Omnidirectional

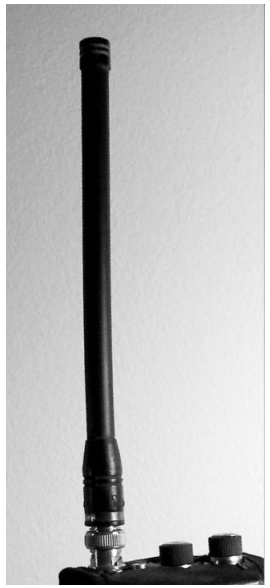


Isotropic
Equal radiation in every direction

Omnidirectional
Uniform in one plane

A dipole still has nulls along its axis.

Wire Antennas



Monopole

Uses a ground plane



Dipole

Two conducting arms



Helical

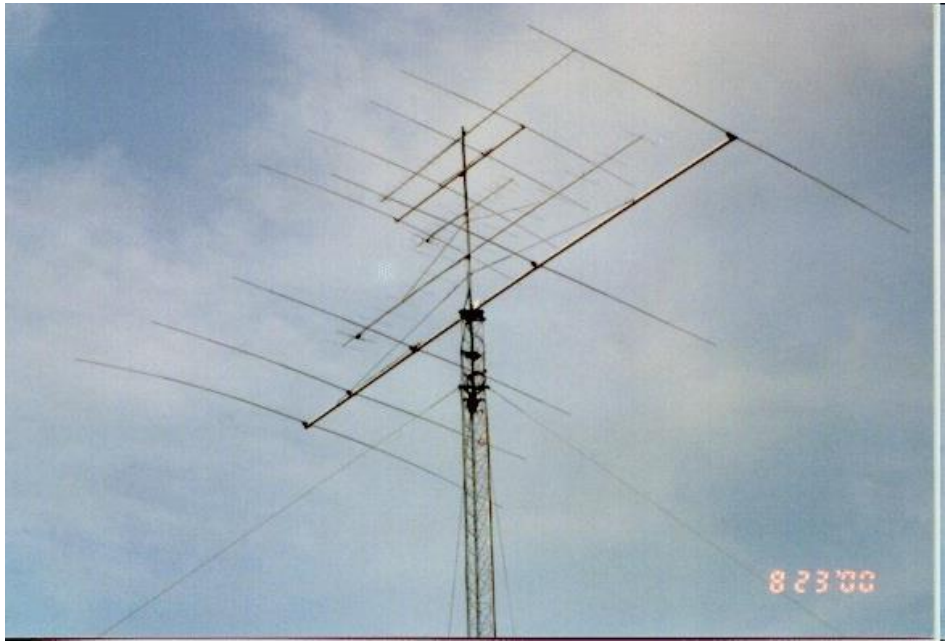
Axial mode can give a directional circularly polarized beam

Log-Periodic and Yagi Antennas



Log-Periodic

Broad frequency coverage



Yagi

Driven element with parasitic elements

Patch Antennas



A radiating patch above a ground plane

Low profile and easy to mount

Can serve as an array element

Horn Antennas

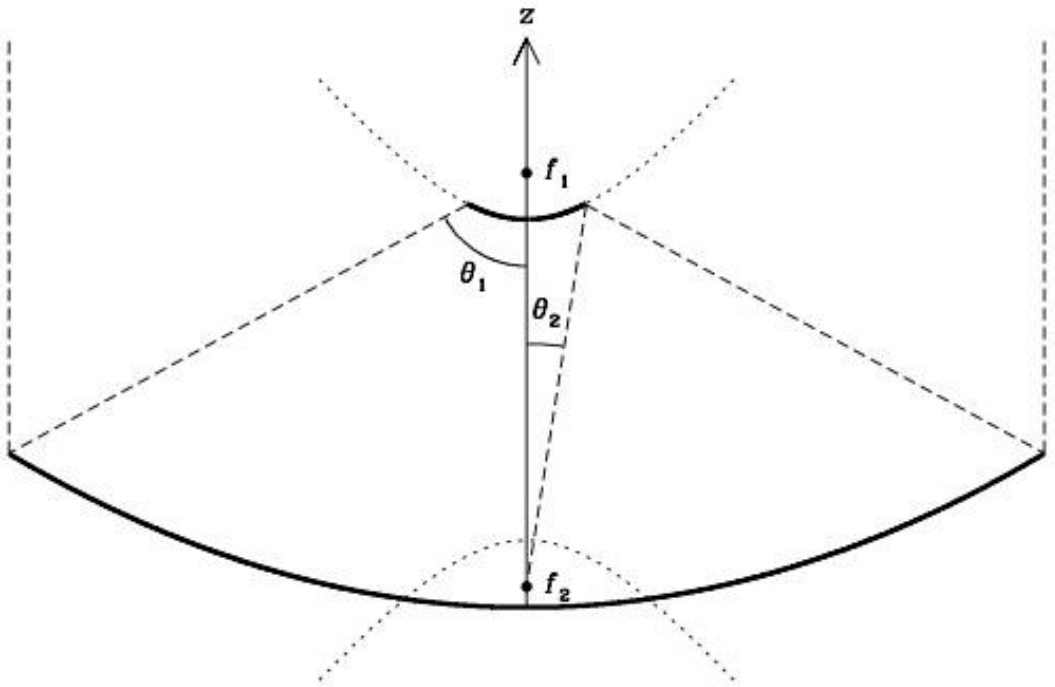


Pyramidal Horns



Conical Horns

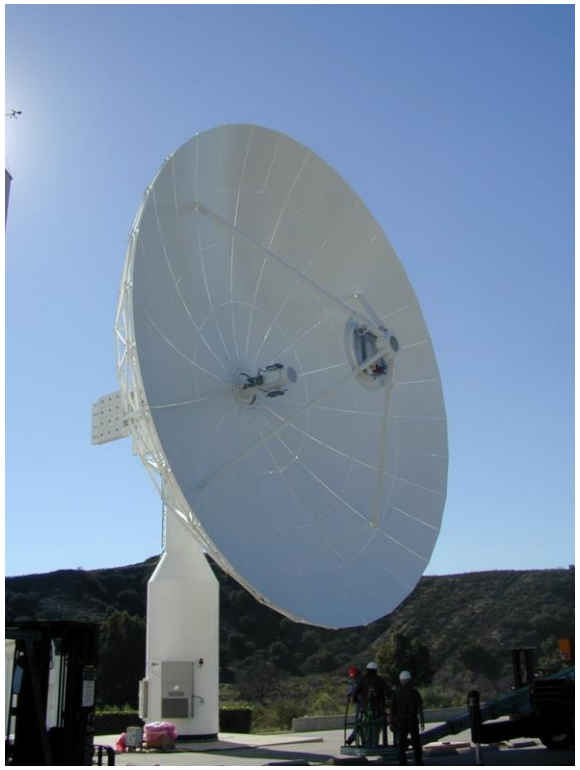
Reflector Configurations



f_1: Prime Focus
Feed at the main reflector's focus

f_2: Cassegrain Focus
A subreflector redirects the path to a feed near the main reflector.

Ground Antenna Hardware

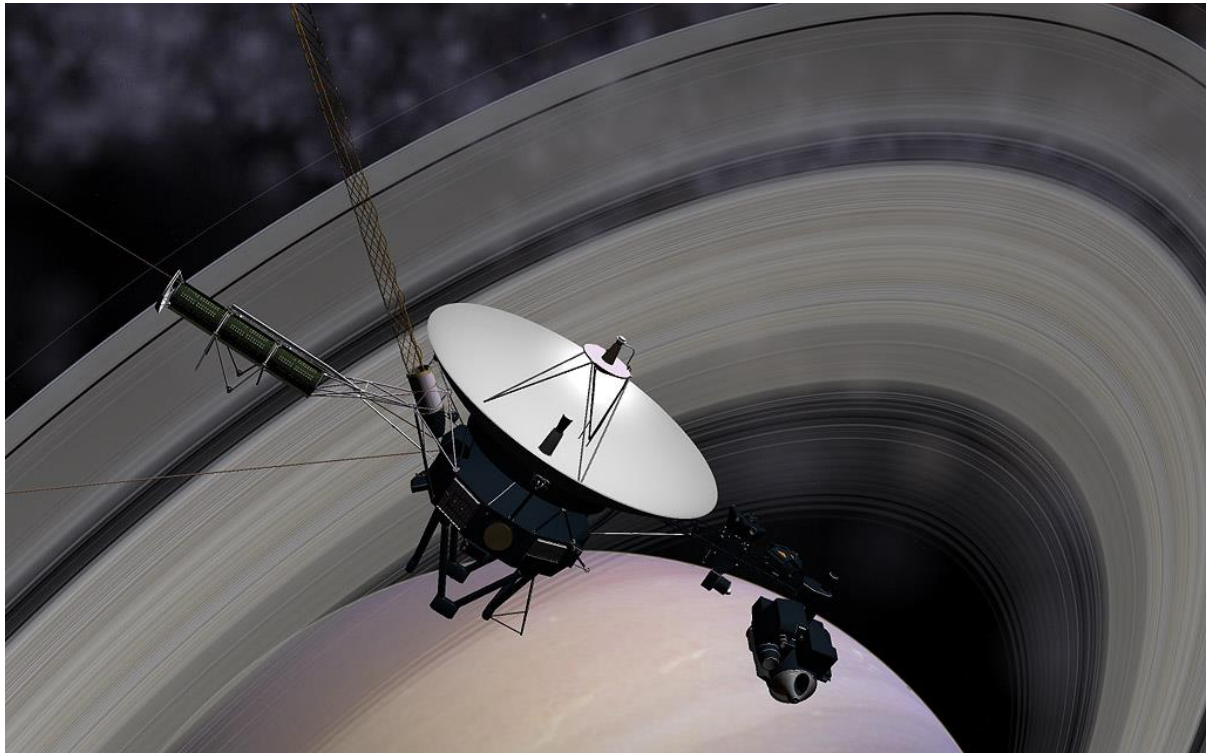


Ground Station Reflector



DSN 70 m Reflector

Antenna Pointing with the Spacecraft



Voyager's high-gain antenna is fixed to the spacecraft.

The spacecraft turns to point the antenna.

What else moves when the antenna moves?

Gimbaled Antennas

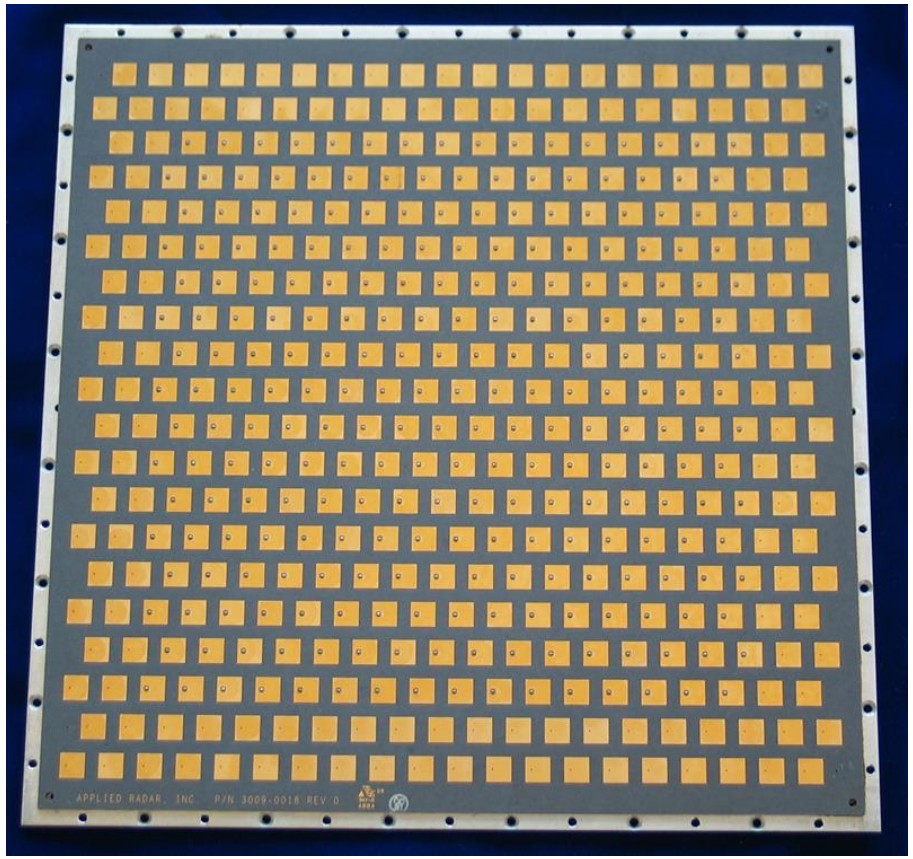


TDRS Reflectors
Boom mounted, two-axis gimbals

Pointing can change without turning the whole spacecraft.

Mechanisms add limits on motion, reliability, and routing.

Phased-Array Hardware



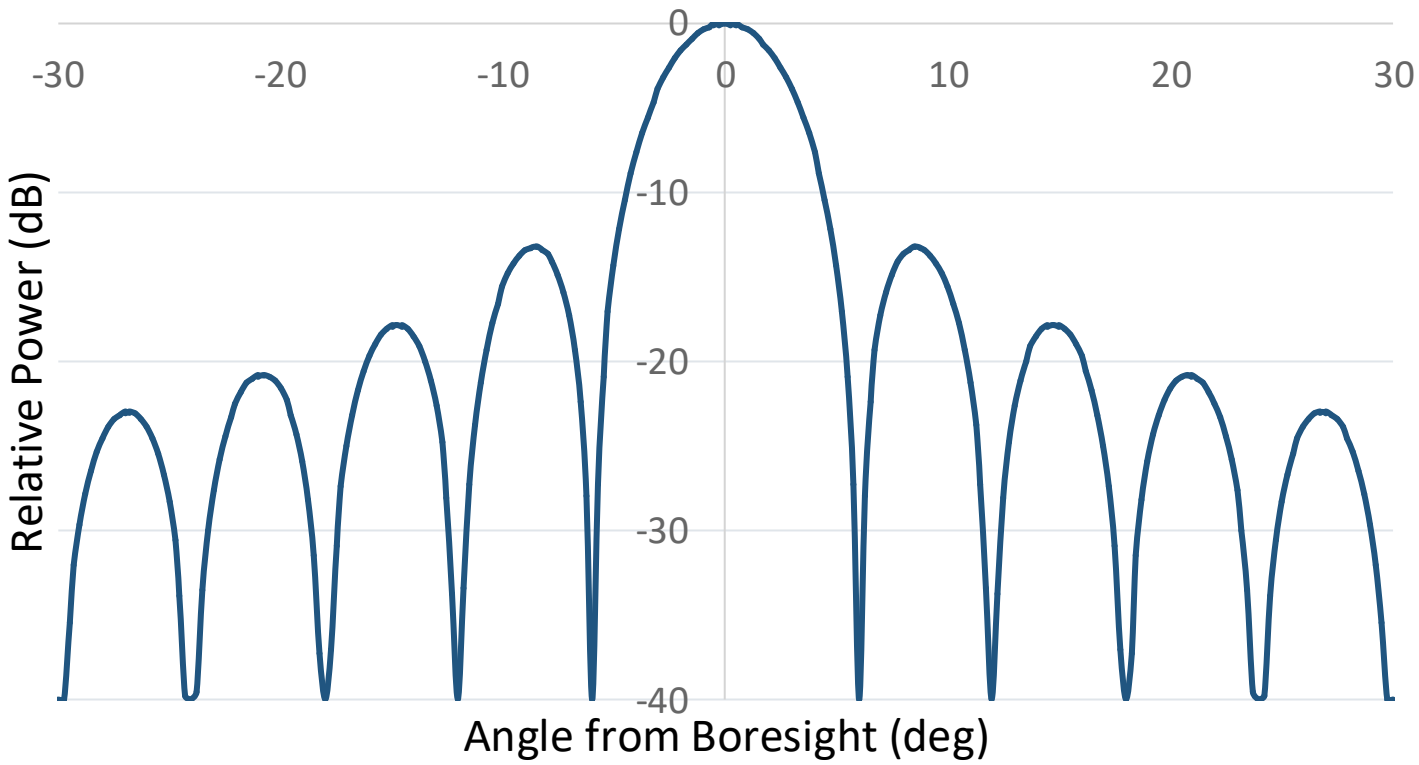
Many elements form one aperture.

Electronics control how their signals combine.

The aperture can remain fixed while the beam moves.

We will return to the steering mechanism in Part 3.

Reading an Antenna Pattern



Boresight
Peak direction

Main Lobe
Near the peak

Sidelobes and Nulls
Other directions

The Receive Antenna

$$P_r = S\,A_e$$

S: Incident Power Flux Density (W/m^2)

A_e: Effective Collecting Area (m^2)

P_r: Available Received Power (W)

Physical Area and Effective Area

For a Circular Aperture: $A = \pi D^2 / 4$

$A_e = \eta_{ap} A$

Use $\eta_{ap} = 0.55$ for our examples.

Illumination, spillover, blockage, and surface accuracy affect how well the physical aperture contributes.

η_{ap} is aperture efficiency. Its meaning differs from radiation efficiency alone [5].

Frequency and Antenna Size

$\lambda \approx 0.300 / f$ f in GHz, λ in m

Frequency	Wavelength	A 1 m Aperture Spans
2.1 GHz	0.143 m	7 Wavelengths
8.4 GHz	0.0357 m	28 Wavelengths

The antenna's size in wavelengths determines much of its behavior.

Gain from Diameter and Frequency

Gain = Effective Area ÷ Isotropic Effective Area

$$[G] = 17.81 + 20 \log_{10}(D) + 20 \log_{10}(f)$$

D in meters. f in GHz. [G] in dBi.

17.81 includes efficiency 0.55 and all constants.

Double Diameter or Frequency: Gain Increases 6.02 dB

Circular aperture; efficiency assumed to be 0.55. All logs are base 10 [4].

A 1 m Antenna at 8.4 GHz

Given: D = 1.00 m f = 8.4 GHz

$$[G] = 17.81 + 20 \log_{10}(D) + 20 \log_{10}(f)$$

Calculate [G] in dBi. Then Predict:

Double D, with frequency fixed: How much does gain change?

Double frequency, with D fixed: How much does gain change?

This gain formula assumes aperture efficiency $\eta_{ap} = 0.55$.

The 1 m Antenna: Results

$[G] = 17.81 + 20 \log_{10}(1.00) + 20 \log_{10}(8.4)$

$[G] = 17.81 + 0.00 + 18.49 = 36.30$ dBi

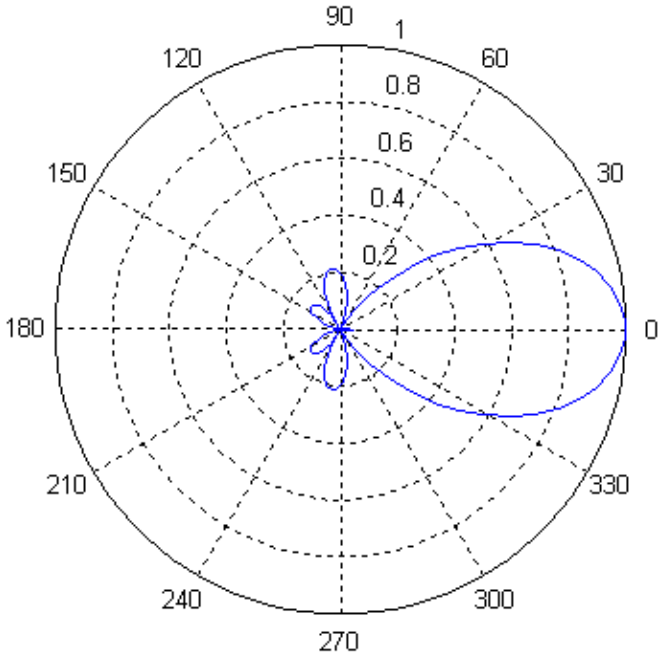
Double D, Fixed Frequency: $\Delta[G] = 20 \log_{10}(2) = +6.02$ dB

Double Frequency, Fixed D: $\Delta[G] = 20 \log_{10}(2) = +6.02$ dB

Double Both D and Frequency: Gain Increases 12.04 dB

D in meters; f in GHz. Efficiency is assumed to be 0.55.

Beamwidth

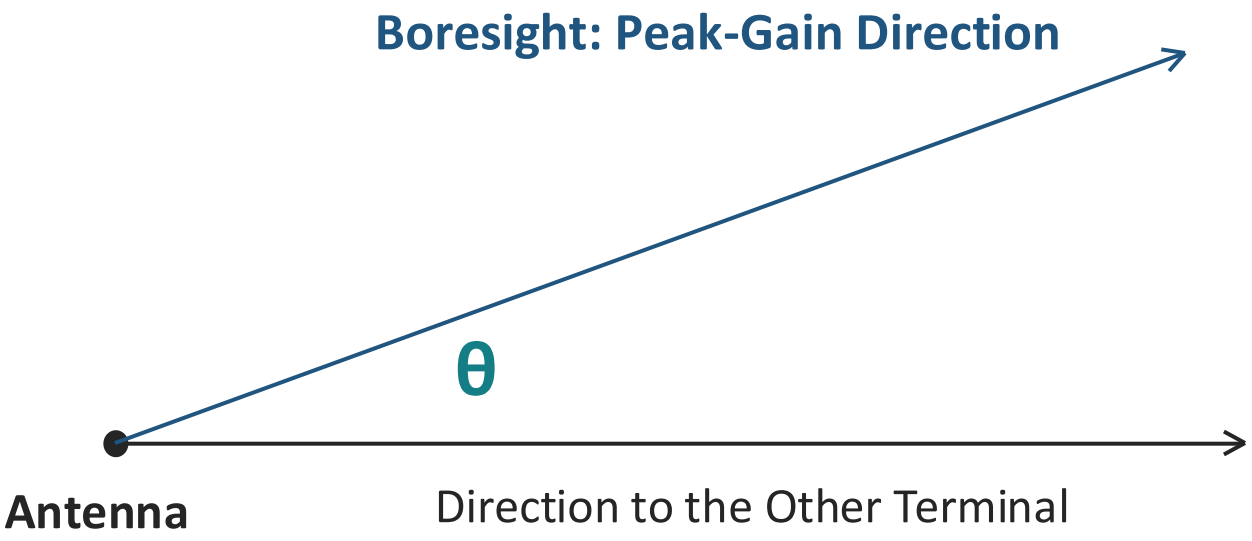


HPBW = Full angle between the two half-power points

Reflector Rule: f in GHz, D in m
HPBW ≈ 21 / (f D) degrees

Approximate; depends on illumination.
1 m at 8.4 GHz: about 2.5°
Half-angle: about 1.25°

What Is Pointing Loss?



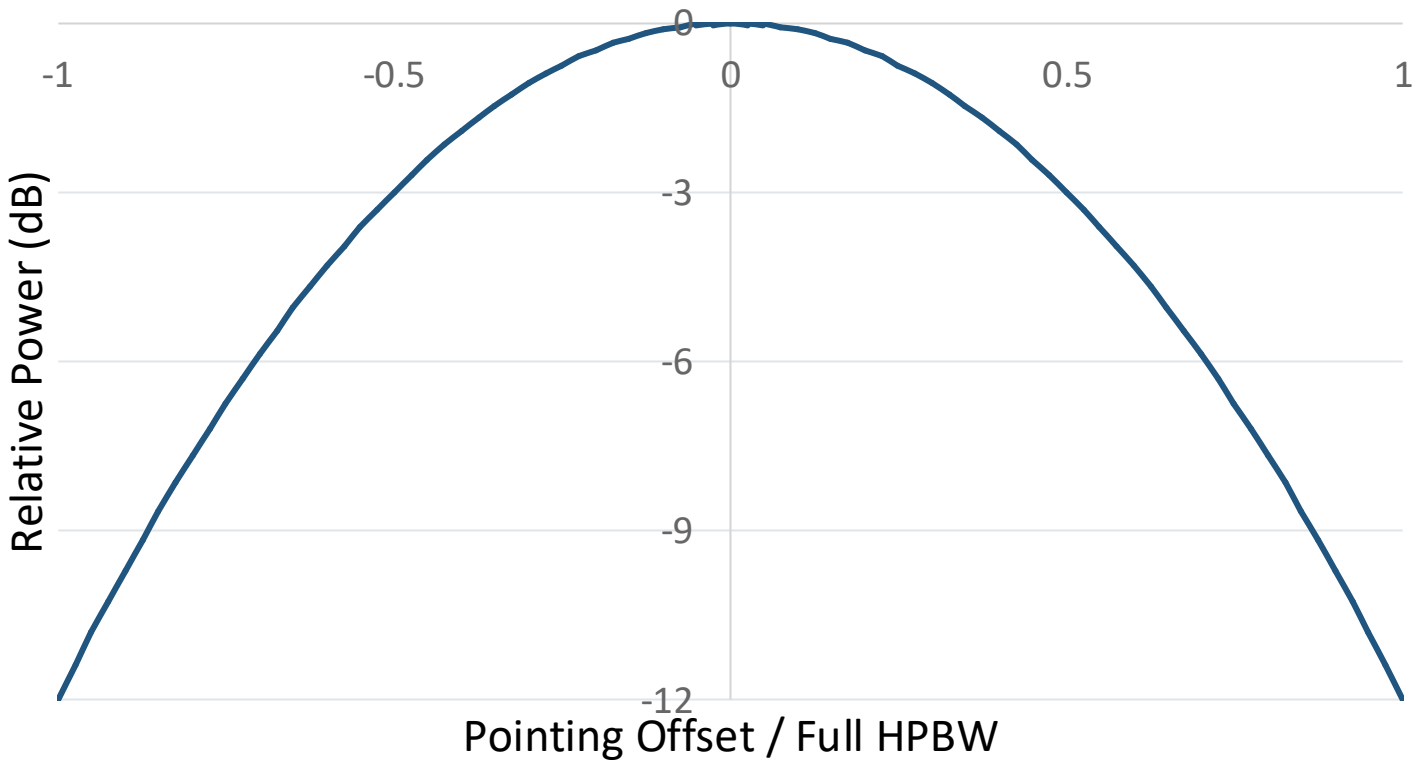
θ is the angle between boresight and the link.

The actual direction has lower gain.

$$[L_{point}] = [G_{peak}] - [G(\theta)]$$

Loss is positive in dB. Evaluate pointing at both ends. The actual pattern may depend on two angles.

Pointing Away from the Peak



Offset / Full HPBW
¼: about 0.75 dB loss
½: about 3 dB loss

At 0.5° error:
2° beam: 0.75 dB
1° beam: 3 dB

Gaussian main-lobe illustration: relative power = −12(offset / HPBW)² dB. Actual patterns differ.

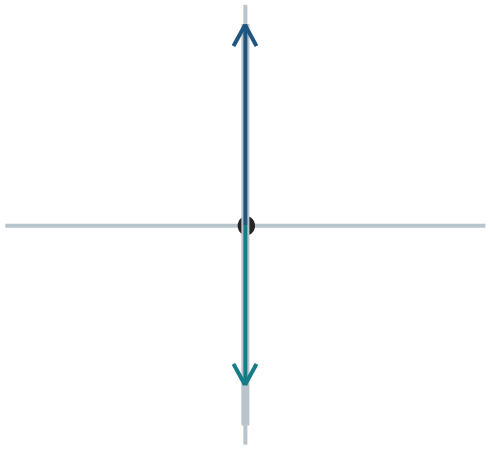
How an Antenna Tracks a Spacecraft

Method	What Drives It?	Practical Issue
Program Track	Predicted position vs. time	Orbit and alignment accuracy
Step Track	Small moves to seek peak signal	Needs usable signal and time
Autotrack	Measured angular error	Special feed/receiver hardware

Prediction can acquire the target before signal-based tracking takes over.

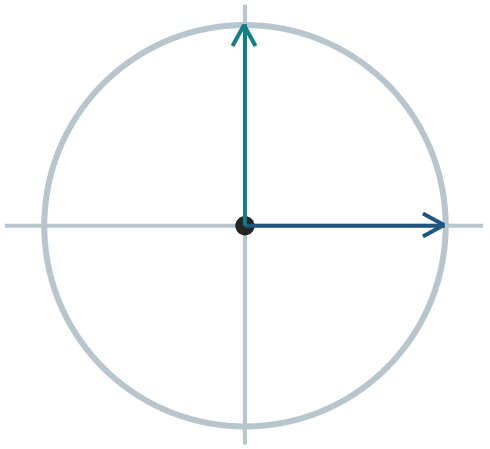
What Is Polarization?

The electric field pushes charges. Its arrow shows direction and strength.
Polarization describes the arrow's motion at one fixed point.



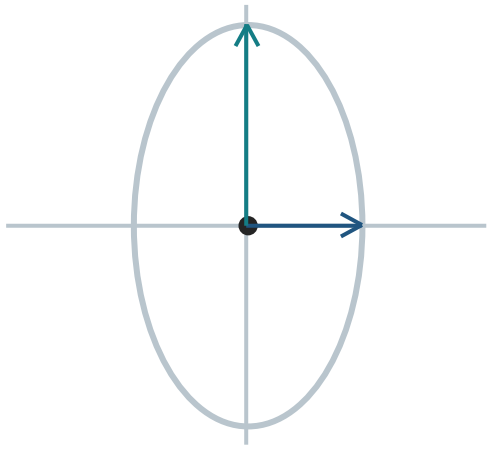
Linear

Back and forth
along a fixed line



Circular

Rotates with
constant length



Elliptical

Rotates with
changing length

End-on view, perpendicular to wave travel. Colors show different instants. Gray traces show the tip's path [6, 11].

Polarization Loss for Linear Antennas

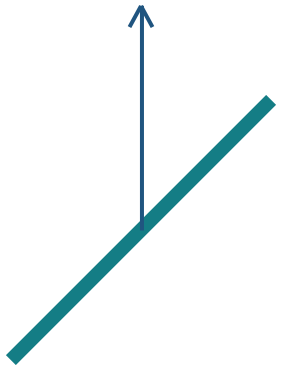
A linear receiver responds to the field component along its polarization axis.



0°

100% Power

0 dB Loss



45°

50% Power

About 3 dB Loss



90°

0% in the Ideal Case

Complete Rejection

Blue: Incident E Field Teal: Receiver Axis

$\eta_{pol} = \cos^2\psi$ $[L_{pol}] = -10 \log_{10}(\eta_{pol})$

ψ is rotation about the line of sight, with pointing fixed. Real antennas have finite isolation [6, 11].

Circular Polarization and Real Antennas

Wave / Receive Polarization	Ideal Coupled Power	Mismatch Loss
Same Circular Sense	100%	0 dB
Opposite Circular Senses	0%	Complete Rejection
Circular / Linear	50%	About 3 dB

Matched circular pairs tolerate rotation about the line of sight.
RHCP / LHCP = Right- / Left-Hand Circular Polarization.

Real polarization is often elliptical. Axial ratio describes its shape.
AR = Major / Minor Field Amplitude. A circle has AR = 1 (0 dB).

Use a consistent viewing convention for handedness. Actual mismatch depends on both polarizations [6, 11].

Pointing and Polarization in the Link Budget

$$[P_r] = [P_{ro}] - [L_{pt}] - [L_{pr}] - [L_{pol}]$$

Illustrative Link	Value
$[P_{ro}]$: Aligned, Polarization-Matched Power	−120 dBW
Pointing Loss: $[L_{pt}]$ Transmit / $[L_{pr}]$ Receive	1 dB / 2 dB
$[L_{pol}]$: Polarization Mismatch Loss	3 dB
Actual Received Power	−126 dBW

For a −125 dBW threshold, the margin is −1 dB.

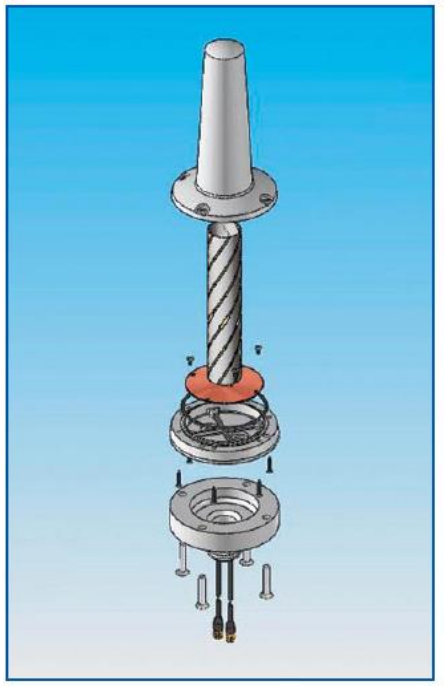
These losses change whether this link supports the selected service.

Aligned power includes other terms. Use peak gains with pointing losses, or actual direction gains without subtracting them again.

Hemispherical Coverage

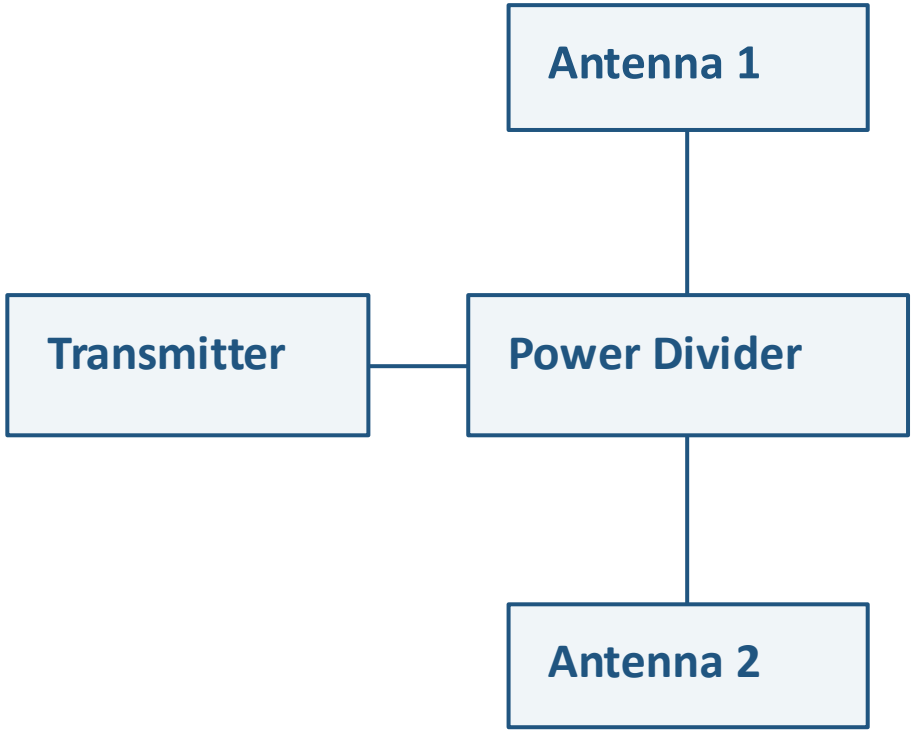


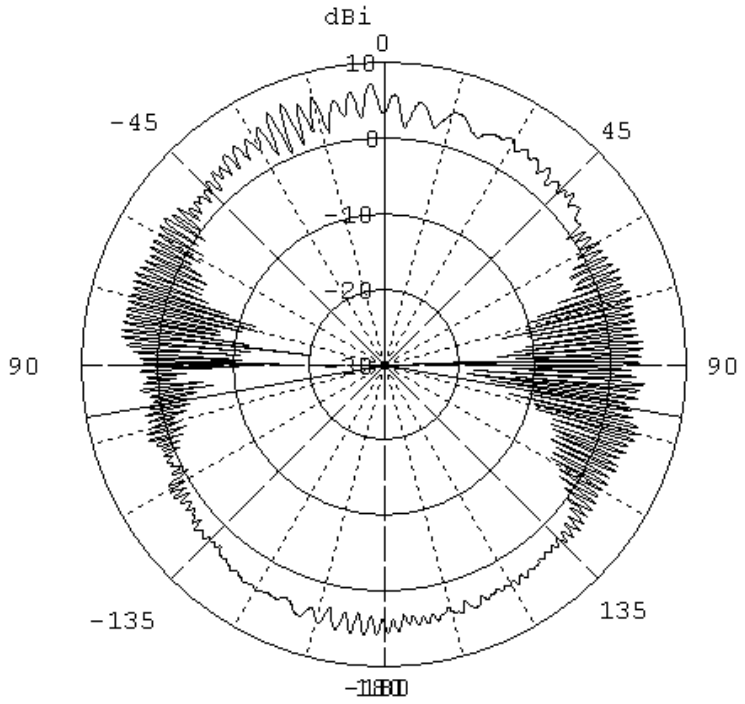
Crossed Dipoles



Quadrifilar Helix

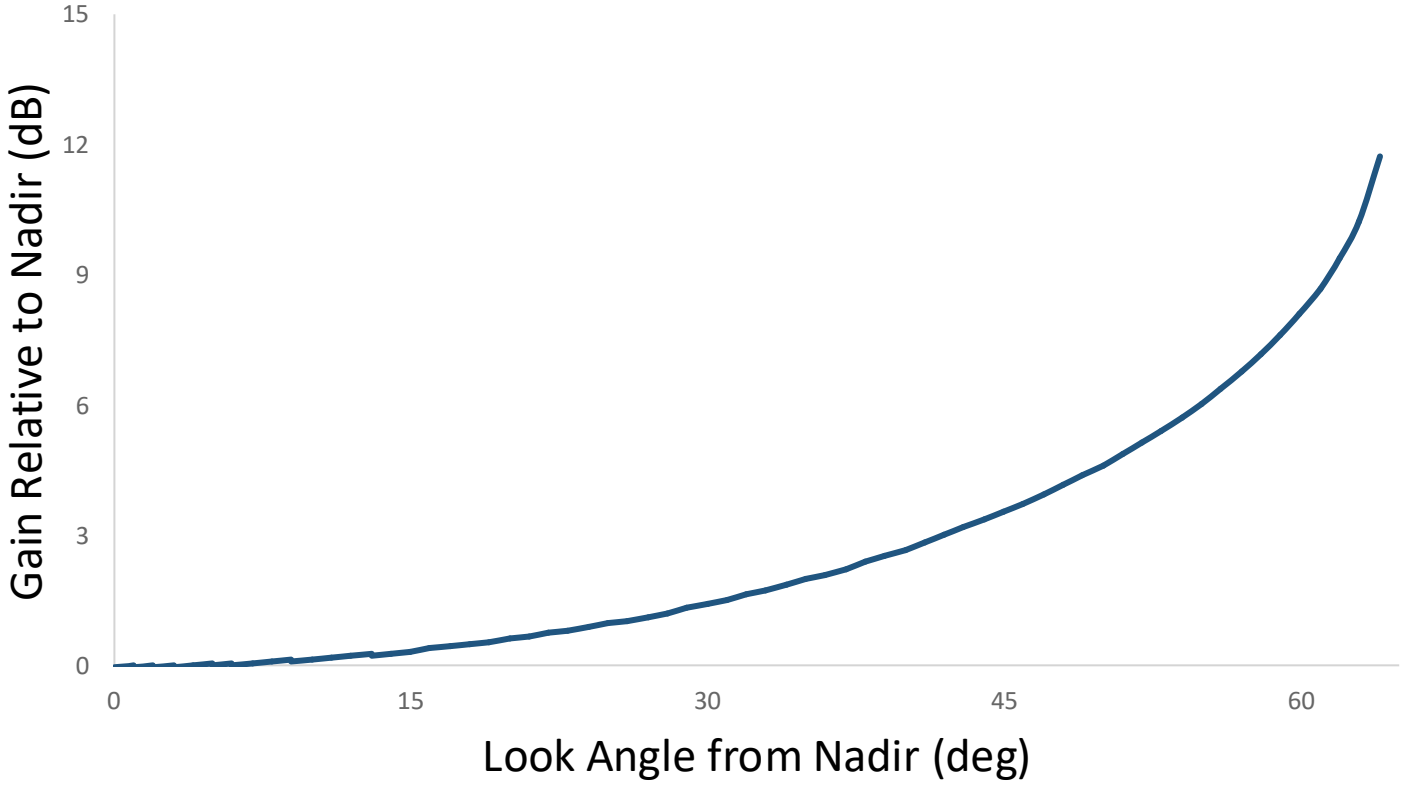
Broad Coverage from Multiple Antennas





Opposite-facing antennas

Shaping Gain for an Earth Pass



Longer slant range near the limb needs more gain.

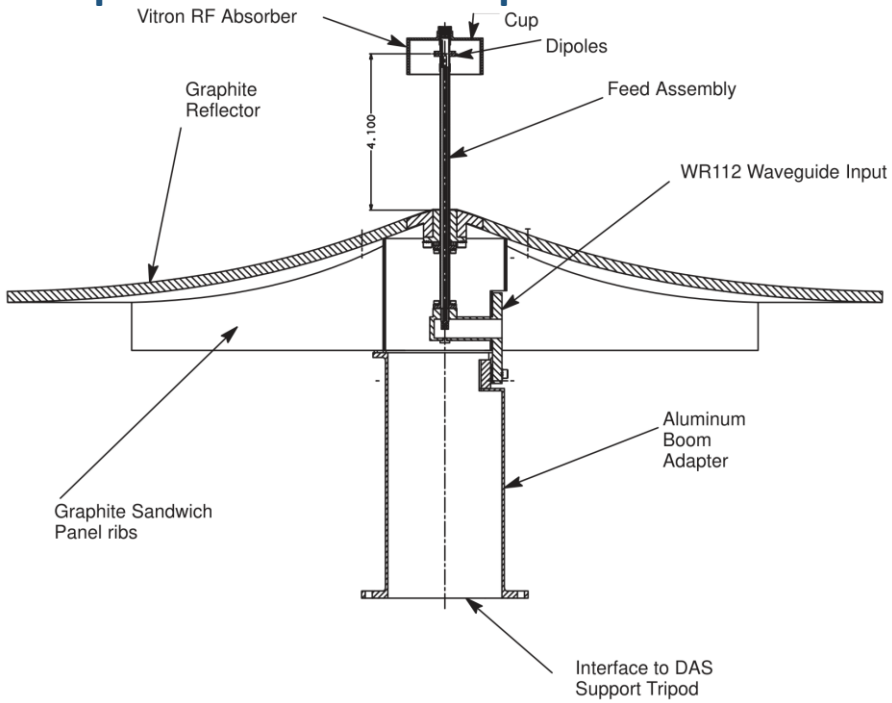
A shaped pattern can reduce the received-power variation.

Assumes nadir pointing.

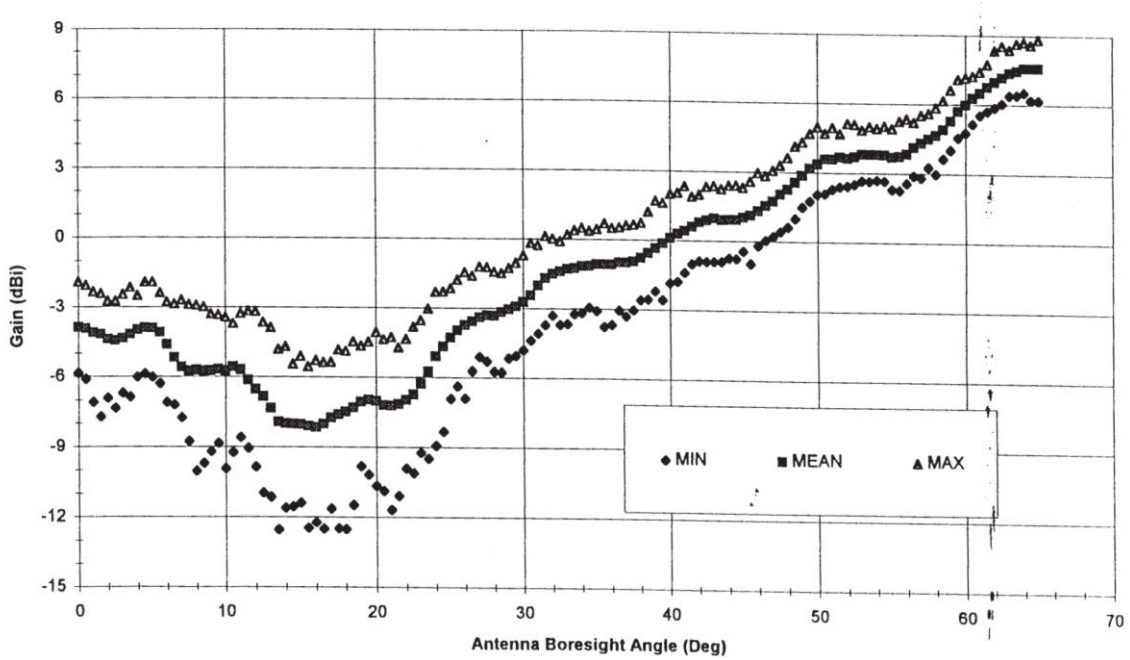
The Earth-Science Shaped Omni

Terra (EOS-AM) Direct Access System, 8.2125 GHz

Shaped Reflector and Dipole Feed



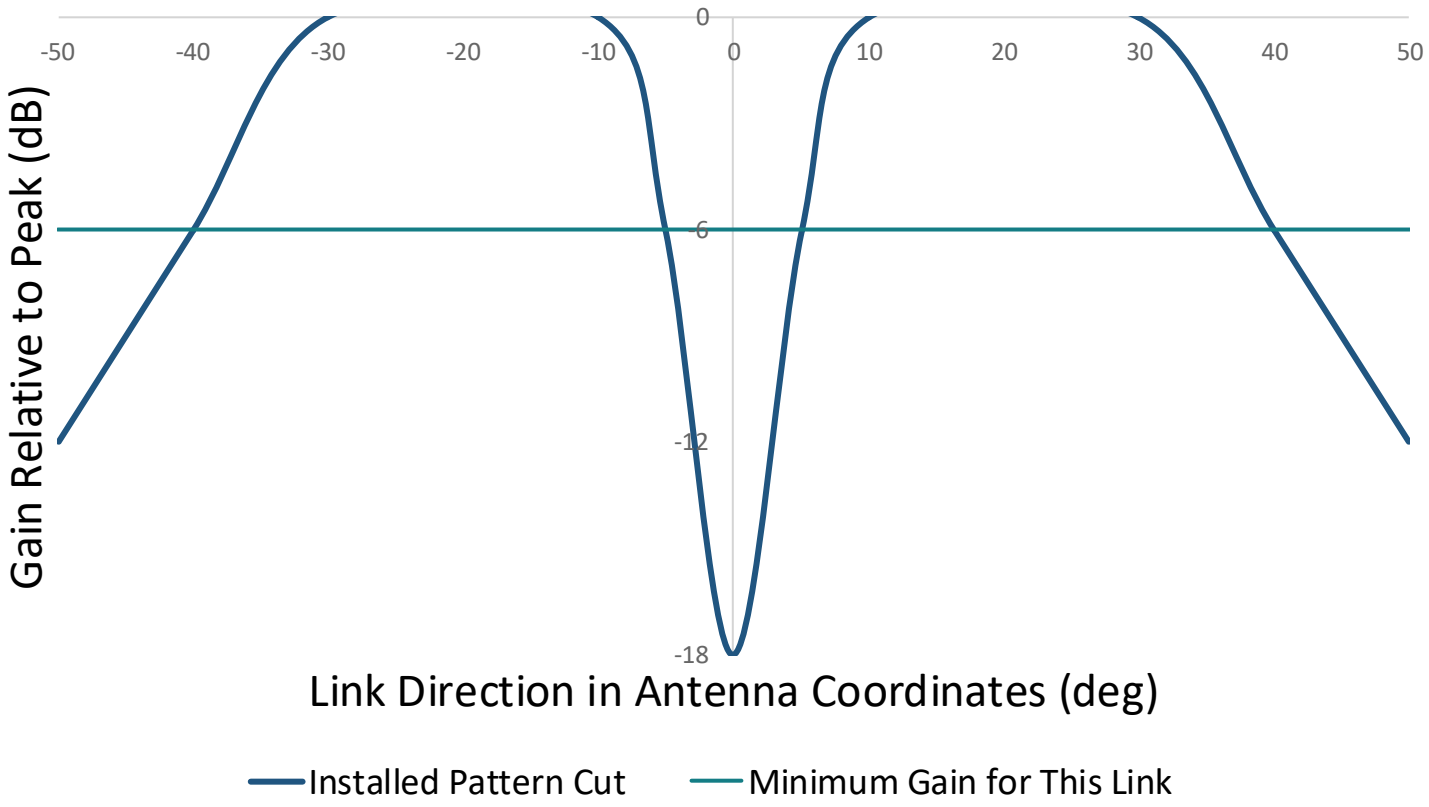
Measured Gain Pattern



More gain toward the limb helps compensate for greater range.

1998 design record. The dip near 20° off nadir also constrained planned direct playback [12, §6, Figs. 2 & 14].

Antenna Patterns and Usable Coverage



The pattern gives gain for each direction.

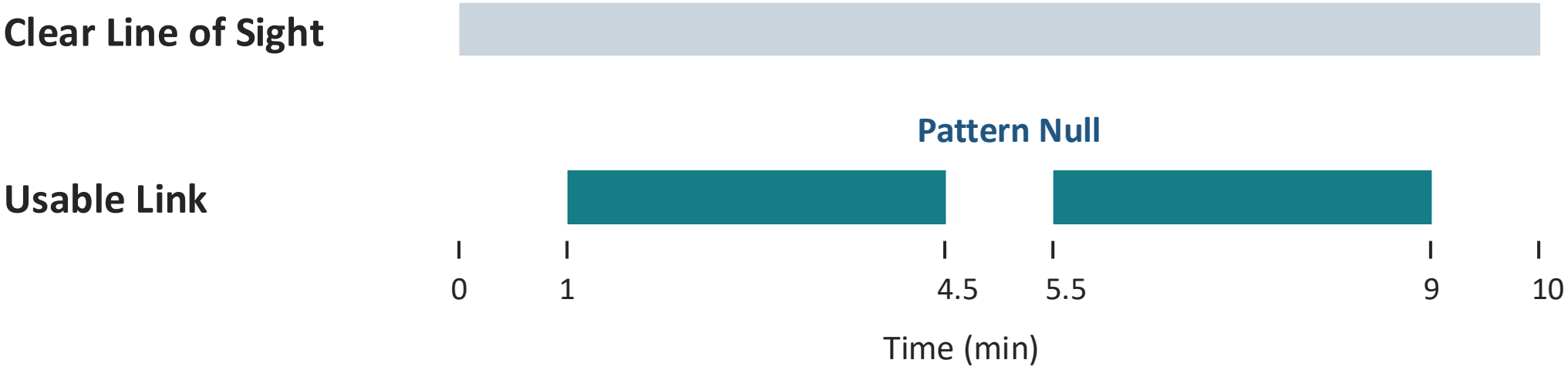
The link budget sets how much gain is enough.

This null creates a gap even with clear line of sight.

Illustrative pattern, not measured hardware. Hold range and other terms fixed. Peak margin = 6 dB. Nulls may arise from interference.

Coverage Becomes Availability over Time

Assume the link direction sweeps from −50° to +50° in 10 minutes.

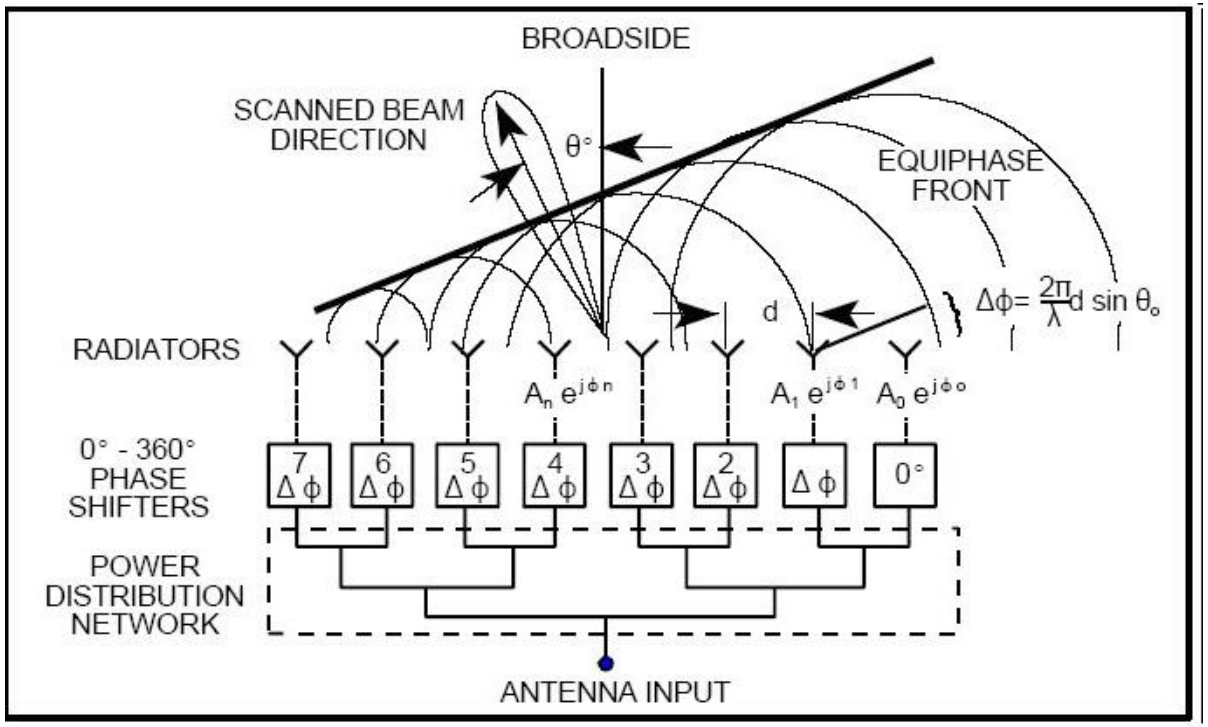


Usable time = 3.5 + 3.5 = 7 minutes

Availability during this visible interval = 7 / 10 = 70%

State the denominator. Mission availability also includes visibility, attitude, tracking, polarization, weather, and scheduled resources.

Electronic Beam Steering



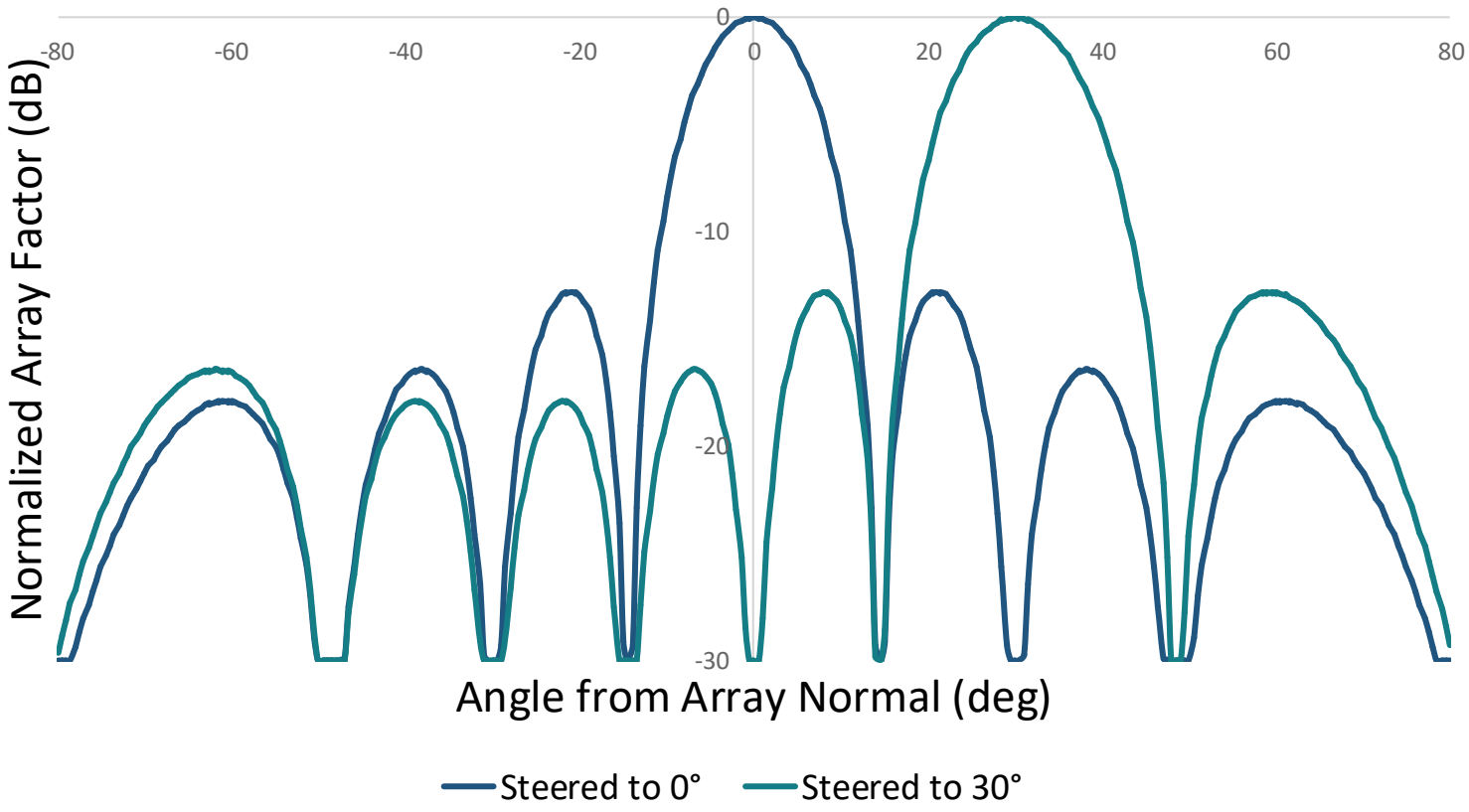
Relative phase or delay sets the direction of constructive combination.

The elements stay fixed.

Transmit and receive use the same principle.

Beam-steering explanation: [10].

The Same Array, a Different Beam



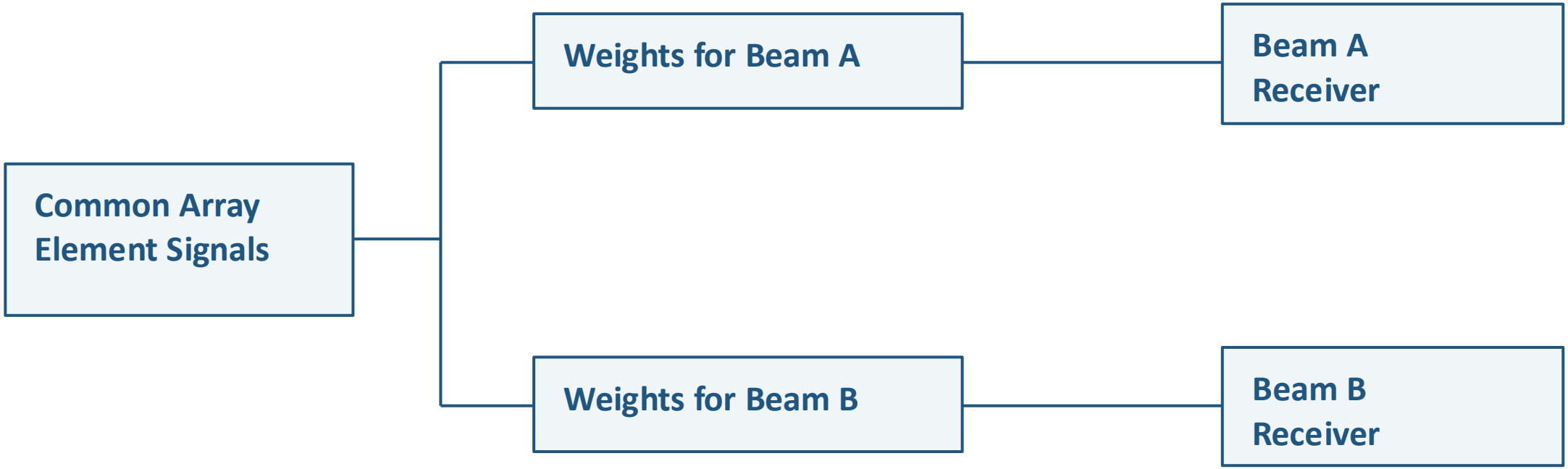
Eight elements
Spacing = λ/2

Change the phase progression to move the peak.

Real scan performance also depends on each element's pattern.

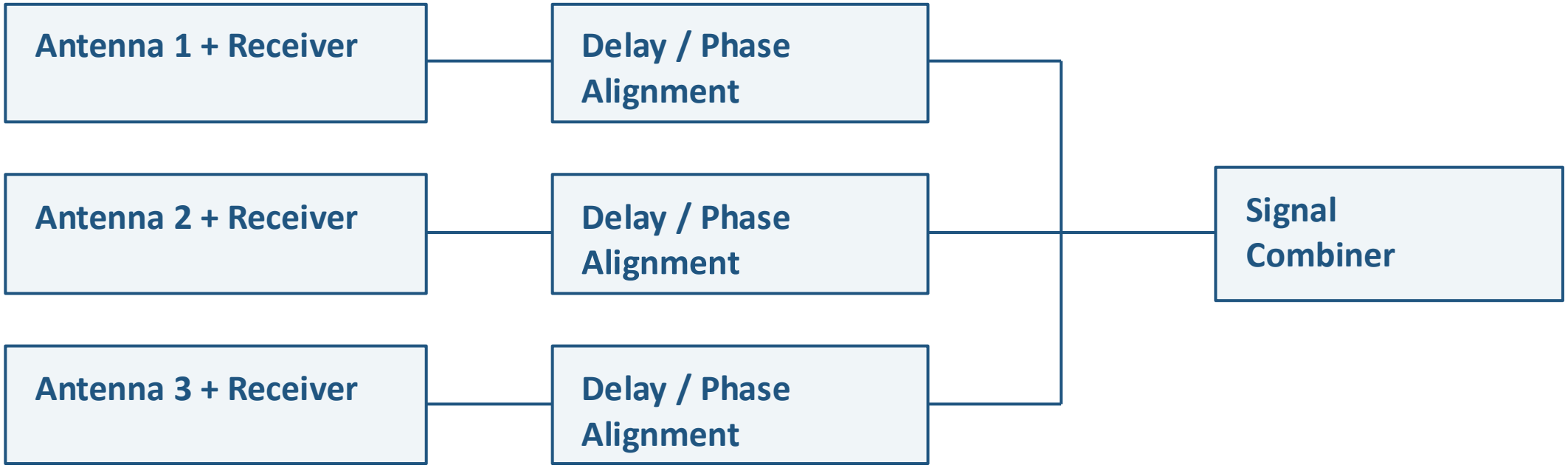
Calculated uniform linear-array factor, equal element amplitudes. Each trace is a separate steering setting [10].

Multiple Beams from One Aperture



Separate combining weights select different directions.

Arraying Ground Antennas



All antennas observe the same spacecraft signal.

Combine aligned voltages or baseband signals. Voyager used ground-antenna arraying [1, sections 6.3.2–6.3.3].

What Arraying Buys Us

Equal Receive Antennas	Ideal SNR Improvement	Equivalent Area
1	Reference	A_e
2	About +3 dB	$2A_e$
4	About +6 dB	$4A_e$

Four equal 34 m apertures collect like one 68 m aperture
if efficiencies match. That does not make their patterns identical.

Assumes equal signals/noise, uncorrelated receiver noise, ideal alignment and combining. SNR = Signal-to-Noise Ratio.

The Very Large Array



Photo: NRAO/AUI/NSF [14]

27 Antennas

25 m Each

New Mexico

Signals Combine for Radio Astronomy

What Changes When Frequency Changes?

$$[P_r] = [P_t] - [L_t] + [G_t] - [L_p] + [G_r] - [L_{other}]$$

Free-Space Path Loss Increases with Frequency.

For Fixed Apertures, Antenna Gains Also Increase.

We must say what stays fixed before comparing links.

The next comparison holds range, transmit power, and other losses constant [4, 5, 7].

Doubling Frequency: Three Comparisons

What Stays Fixed?	$\Delta[G_t]$	$\Delta[G_r]$	$\Delta[L_p]$	$\Delta[P_r]$
Both Antenna Gains	0	0	+6 dB	−6 dB
Tx Gain and Rx Aperture	0	+6 dB	+6 dB	0
Both Physical Apertures	+6 dB	+6 dB	+6 dB	+6 dB

For fixed apertures, assume unchanged efficiency and accurate pointing.

Rounded changes. Free-space comparison with constant transmit power and other losses [4, 5, 7].

Fixed Apertures at Two Frequencies

Quantity	2.1 GHz	8.4 GHz
Tx Gain: 1 m Aperture	24.25 dBi	36.30 dBi
Rx Gain: 3 m Aperture	33.80 dBi	45.84 dBi
Path Loss: 2,000,000 m	164.91 dB	176.96 dB
Received Power: 10 W Tx	−96.86 dBW	−84.82 dBW

Four Times the Frequency: About 12 dB More Received Power Here

η_{ap} = 0.55. Use gain constant 17.81; path-loss constant 32.45 with R in m, f in GHz.

Frequency Choices in a Real Mission

Pointing and Surface Accuracy Become More Demanding as frequency increases.

Atmospheric Attenuation Depends on Frequency and Weather.

Available Spectrum and Compatible Ground Equipment also constrain the choice.

The free-space calculation isolates one part of this trade [5, 8].

Antenna Selection Choices

Choice	Examples	What It Affects
Radiating Structure	Wire, patch, horn, reflector	Pattern, gain, bandwidth
Coverage Control	Body, gimbal, beamforming	Pointing, power, complexity
Deployment	Fixed, hinged, unfurlable	Mass, packaging, reliability
Feed and Polarization	Single/multiple feeds, linear/circular	Losses and compatibility

Antenna Trades Across the Spacecraft

Mass: Aperture + Supports + Deployment + Pointing Hardware

Attitude Control: Can We Keep the Beam on the Ground Station?

Structures: Can We Package, Deploy, and Support the Antenna?

Power and Thermal: What Does the Chosen Architecture Require?

Operations: What Link Is Available During Acquisition or Safe Mode?

Which trade would matter most for a small Earth orbiter? For a distant planetary mission?

Antenna Choices for the Mission

The ground station can see the spacecraft, but the link drops.
What would you check besides range?

Would a broader beam help? What would it cost?

Would circular polarization help if the spacecraft rolls?
What must still match at the receiver?

Possible checks: installed pattern, attitude, pointing, polarization, receiver threshold, and resource availability.

What We Can Now Explain

Pointing loss comes from gain in the actual link direction.

Polarization loss comes from incomplete field coupling.

A pattern and a link threshold define usable directions.

Motion through those directions determines usable time.

Orbit and attitude connect antenna coordinates to mission coverage. Availability always needs a defined time interval.

Before We Finish

A larger antenna gives us more gain.
What might prevent us from using it?

Which part of today's calculation would you like to revisit?

Homework Questions and Next Steps

Optional: The Far-Field Model

For an Electrically Large Aperture: $R \gtrapprox 6.67\, D^2\, f$

D and R in meters, f in GHz. Also require $R \gg D$ and $R \gg 0.300/f$.

At 8.4 GHz: 3 m Aperture: About 504 m
70 m Aperture: About 274,000 m

Additional References

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Optional references. Numerical antenna examples and pattern plots are teaching calculations.